Musa Buğra Demir

Computer Engineering Student · Robotics, Embedded & AI Software · Game Designer

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GitHub: github.com/NightKnight-nx2 **LinkedIn:** linkedin.com/in/musabugrademir **Nationality:** Turkish

ABOUT ME

Third-year Computer Engineering student at Ege University and Chair of the IEEE Ege Student Branch (600 members). I build control, perception and communication software for autonomous vehicles, leading software and communications on five Teknofest teams. Software lead of the IEEE Region 8 Student Hardware Contest world top 20 team, 3rd in Türkiye at IEEEXtreme 20.0, and lead game designer of a Steam title with 10K sales.

EDUCATION AND TRAINING

09/2023 - CURRENT · İzmir, Türkiye
B.Sc. Computer Engineering · Ege University
EQF level 6 · ege.edu.tr

WORK AND LEADERSHIP EXPERIENCE

2026 - CURRENT · İzmir, Türkiye
Chair of the Executive Board · IEEE Ege Student Branch, Ege University

- Set a term goal of growing the branch from 7 to 30 active projects; reached 20 projects so far and growing.
- Grew membership to 600 after a low-activity previous year.
- Built and run an active sponsorship network that continues to secure agreements with major companies.
- Enforce an open-source output policy: the code of every branch project is published on GitHub.

06/2025 - 2026 · İzmir, Türkiye
Treasurer and Board Member · IEEE Ege Student Branch, Ege University

- Ran financial planning, budgeting and reporting; worked with the sponsorship and event teams to fund Butterfly Effect, SciTech '26, İzmir Energy Summit, RAStlantı and CyberEge.
- Headed the organisation and coordination teams of İş Hayatında Kelebek Etkisi'9 (03/2026) and SciTech (02/2026); coordination team member at GreenInvest (EU-supported), CyberEge24 and the IEEE Aegean Regional Meeting.
- Board member responsible for Teknofest projects and the CS, ComSoc, WIE and Promotion-Design committees; chaired the Communications Society chapter.
- Designed and built the branch website and admin panel, ieeeegesb.org, end to end: event registration, QR ticket scanning, certificate issuing and verification, leaderboard, consent management and audit logging (26 data models, 80+ automated tests).

Stack: React, Vite, Node.js, Express, TypeScript, PostgreSQL, Prisma, Redis, Playwright

2025 - CURRENT · İzmir, Türkiye
Software & Communications Captain, Teknofest and Hardware Teams · IEEE Ege Student Branch

Mavi İnci, Unmanned Surface Vehicle · Software Captain 2025-26, Software & Comms Captain 2026 - Current

- Main developer of a 15-package ROS 2 autonomy stack: YOLOv8 buoy detection on the TensorRT C++ API, Nav2 navigation, a P1/P2/P3 mission state machine, an FDIR safety layer with remote e-stop, and a MAVLink/MAVROS bridge.
- Kept one codebase for simulation (Gazebo Harmonic, ArduPilot SITL) and the real vehicle (Jetson, Pixhawk); pool-tested the vehicle in July 2026.

AquaVena, EGEduino Team · Software Lead · IEEE Region 8 Student Hardware Contest 2026

- Smart IV infusion module placed in the world top 20. Firmware on ESP32-S3 with ESP-IDF and FreeRTOS: 11,523 hand-written lines of C, 9 tasks, 17 LVGL screens, clean build with 0 errors and 0 warnings.
- Pinned safety, watchdog and PID to a dedicated core so UI and networking can never delay the 1 ms safety poll; isolated and fixed three stacked faults on a shared SPI bus (display, touch, RFID).

Çelikgöz Çelikkubbe, Air Defence Systems · Software & Comms Captain, 2025 - Current

- Leading fire-control and effective-range software; trained YOLO models with TensorRT for real-time target detection and tracking; built C++ command-and-control links.

ODBARS, Unmanned Ground Vehicle · Communications Captain, 2025-26

- Designed long-range LoRa communication and telemetry for low-latency remote operation; mentored the software team on link reliability and protocol design.

International UAV and Model Satellite (new teams) · Software & Comms Captain, 2026 - Current

- Founding software and communications captain; the Model Satellite team was formed from the former IEEE Ege CubeSat team, where I was Software Team Lead and Project Manager (2025), covering orbit calculation and satellite communication.

TerraSync · Project Lead & Captain, 09/2026 - Current

- Open-source decision-support system fusing Sentinel-2/Landsat imagery with low-cost ESP32 ground sensors to map urban heat islands, green-space loss and water stress at neighbourhood scale; Bornova/İzmir pilot.

Stack: ROS 2, Nav2, C++, Python, TensorRT, YOLO, OpenCV, PyTorch, MAVROS, Gazebo, ESP-IDF, FreeRTOS, LoRa, Google Earth Engine

11/2025 - CURRENT · İzmir, Türkiye

Game Designer and Level Designer · Aegean Crafters (independent game studio)

- **Dual Protocol: The Case of Eva & Miles** (Steam, 10/2026, published by Rock Paper Publisher): two-player asymmetric noir puzzle game; 80K wishlists and 10K sales. Lead game design, level and puzzle design, story and narrative, localization and Steam integration.
- **Spacey Space Rush** (cancelled by the publisher): built online systems, the co-op gameplay loop and a competitive online mode for a 2-8 player party game; level, game and mechanics design.
- **Spacey Space Burger** (on hold): game, mechanics and level design and 3D physics systems for an open-world management sim.

Stack: Unity, C#, Steamworks

SELECTED PROJECTS

- **7-DOF Vehicle Dynamics + Unity Replay:** from-scratch Pacejka tyre model, three driver profiles and pure-pursuit tracking; peak lateral force within 0.1% of a reference library (4200 N vs 4196 N). Python, SciPy, Unity.
- **LunarLander REINFORCE:** 212-parameter policy that escapes a hovering local optimum through entropy annealing; landing rate raised from 3.0% to 91.1% over 3 seeds. Python, PyTorch, Gymnasium.
- **Wheeled-Biped Robot:** Lagrangian dynamics with KKT wheel-ground contact; LQR vs constrained MPC over rough terrain. Python, SymPy, do-mpc.
- **Perception-Control Loop:** Unity front camera with scene-generated labels; YOLOX-nano trained on the synthetic data (AP50 0.917) drives a closed-loop Stanley controller from detected cones, adding only 0.17 s per lap with 59 ms latency. Python, PyTorch, Unity.
- **FSOCO Cone Detection:** YOLO11n on real Formula Student photos with a leak-free, team-based test split: mAP50 0.787 at 4.8 ms per image; ByteTrack tracking and centreline estimation.

More at github.com/NightKnight-nx2 (45+ public repositories).

HONOURS AND AWARDS

2026 **World Top 20,** IEEE Region 8 Student Hardware Contest (AquaVena, team EGEduino, software lead)

2026 **3rd in Türkiye, 32nd worldwide,** IEEEXtreme 20.0, 24-hour global programming contest, out of about 9,000 teams

2026 **"Best Idea" and "Most Sustainable Idea",** Monad Blitz Hackathon (Flutter client, Solidity smart contracts on Monad)

2025 **33rd in Türkiye, 1071st worldwide,** IEEEXtreme 19.0

DIGITAL SKILLS

Languages: Python, C++, C, C#, Java, JavaScript, TypeScript, Dart, Solidity, SQL

AI & Computer Vision: PyTorch, YOLO, TensorRT, OpenCV, MediaPipe, reinforcement learning

Web: React, Vite, Node.js, Express, PostgreSQL, Prisma, Redis

Games & 3D: Unity, Steamworks, Unreal Engine, Blender, SolidWorks

Robotics & Embedded: ROS 2, Nav2, MAVROS, Gazebo, ArduPilot SITL, ESP-IDF, FreeRTOS, LVGL, STM32, ESP32, Arduino, LoRa

Control & Simulation: LQR, MPC, PID, Kalman filtering, Pacejka, SciPy, SymPy

Remote Sensing: Sentinel-2, Landsat, Google Earth Engine, Rasterio, Dask

Tools: Git, Docker, Linux, Playwright, Figma, Photoshop

LANGUAGE SKILLS

Mother tongue: Turkish

	Understanding		Speaking		Writing
	Listening	Reading	Spoken production	Spoken interaction	
English	C1	C1	B2	B1	B2
Spanish	A2	A2	A2	A2	A2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

VOLUNTEERING

STEM Volunteer · IEEE Ege Student Branch

- Children in Power (200+ students), Hour of Code (150+), Teach and Make (100+), From Academia to High School: Women Engineers, and the STAR Project.